

Electronic structure calculations on a fault tolerant quantum computer

IEEE Quantum Week 2026



QUANTUM
MOTION



Session 2: Mapping from chemistry to quantum computers



Tutorial roadmap

Room 717B

Tuesday September 15, 2026

10.00am

10.45am

11.30pm

1.00pm

1.45pm

2.30pm

1 Intro to
quantum
chemistry

2 From
chemistry to
quantum
computers

Lunch

3 Circuit
optimisations
and resource
estimates

4 Automating
resource
estimation on
software

Takeaway from Session 1

We want to find the lowest energy eigenstate of the electronic Hamiltonian:

$$\hat{H}_{el} = -\frac{1}{2} \sum_i \Delta_i + \sum_{\{i,j>i\}} \frac{1}{|r_i - r_j|} - \sum_{\{i,I\}} \frac{Z_I}{|R_I - r_i|} + \sum_{\{I,J>I\}} \frac{Z_I Z_J}{|R_I - R_J|}$$

Approach:

- Introduce a finite basis (e.g. STO-3G, 6-31G*)
- Run Hartree-Fock calculation
 - Obtain a collection of optimised molecular orbitals
 - Hartree-Fock state is the lowest-energy Slater determinant
 - Hartree-Fock energy is different from true ground state energy due to electron correlation
- We need high-accuracy methods, which consider a superposition over multiple Slater determinants
 - The full space of Slater determinants (full-CI) is combinatorially large!

This session: map large Hilbert space onto a quantum computer

Two ways to write down a fermionic state

First quantisation (track electrons)



antisymmetry imposed by hand

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_\eta)$$

Second quantisation (track occupations)



antisymmetry built into the operators

$$|n_0 n_1 \dots n_{N-1}\rangle, \quad n_p \in \{0, 1\}$$

- First quantisation tracks electrons over a basis (e.g., position, molecular orbitals)
- Second quantisation tracks occupation of spin orbitals wrapped up into Slater determinants

Both can be mapped to a qubit-based quantum computer. We'll focus on second quantisation for simplicity

Creation and annihilation operators

Impose anticommutation relations

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \quad \{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0$$

Define occupation-number states

$$|n_0 n_1 \dots n_{N-1}\rangle \equiv (a_0^\dagger)^{n_0} (a_1^\dagger)^{n_1} \dots (a_{N-1}^\dagger)^{n_{N-1}} |0\rangle.$$

$$a_p^\dagger |\dots 0_p \dots\rangle = (-1)^{\sum_{q < p} n_q} |\dots 1_p \dots\rangle \quad \text{Antisymmetry}$$

$$a_p^\dagger |\dots 1_p \dots\rangle = 0 \quad \text{Pauli exclusion principle}$$

-
- Pauli exclusion and antisymmetry are fundamental identities rather than constraints imposed later
 - **The sign factor is important later for mapping to qubits!**

The second-quantised electronic Hamiltonian

First-quantised
electronic Hamiltonian:

$$\hat{H}_{el} = -\frac{1}{2} \sum_i \Delta_i + \sum_{\{i,j>i\}} \frac{1}{|r_i - r_j|} - \sum_{\{i,I\}} \frac{Z_I}{|R_I - r_i|} + \sum_{\{I,J>I\}} \frac{Z_I Z_J}{|R_I - R_J|}$$

Second-quantised
electronic Hamiltonian:

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s + E_{\text{nuc}}$$

↖ Fixed nuclear term

“One-body” terms: electron kinetic energy
and electron-nuclear potential energy

“Two-body” terms: electron-electron
potential and particle exchange

$$h_{pq} = \int dr \phi_p^*(r) \left(-\frac{1}{2} \Delta - \sum_I \frac{Z_I}{|r - R_I|} \right) \phi_q(r) \quad h_{pqrs} = \iint dr_1 dr_2 \frac{\phi_p^*(r_1) \phi_q^*(r_2) \phi_r(r_2) \phi_s(r_1)}{|r_1 - r_2|}$$

These terms depend only on molecular orbitals and can be calculated in advance of full-CI calculations



Fixing an example: the hydrogen molecule H_2

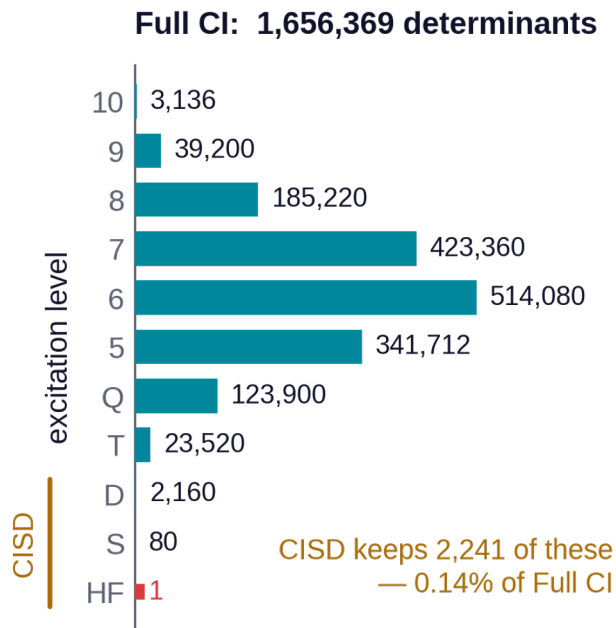
Details: STO-3G basis, $R = 0.7414 \text{ \AA}$, 2 spatial orbitals = 4 spin-orbitals, 2 electrons (all energies in Ha)

one-body terms		two-body terms	
h_{00}	-1.25246357	(00 00)	0.67448877
h_{11}	-0.47594872	(00 11)	0.66346810
		(01 01)	0.18128881
		(11 11)	0.69739377

E_{nuc}	E_{HF}	E_{FCI}	correlation
0.71375399	-1.11668439	-1.13727017	-0.02058579

This afternoon, Vasco will show how to obtain these input numbers using ORCA

Compare to a scaled-up example



H₂O / 6-31G · 13 spatial orbitals = 26 spin-orbitals
10 electrons: 5 ↑ + 5 ↓

Requirements for full-CI

Classical: keep track of 1.6M floats describing the superposition

Quantum: use 26 qubits to represent the occupation of each spin orbital

The rest of this talk:

How do we put fermionic creation and annihilation operators onto qubits?

How do we extract the energy back out?



Fermions to qubits

We can naively assign one qubit to each spin orbital, but how should we map our fermionic operators to qubit operators?

Fermionic input

Operators are anticommuting:

$$\{a_i^\dagger, a_j^\dagger\} = 0$$

Qubit output

Pauli operators are commuting ($i \neq j$):

$$[X_i, X_j] = 0$$

Jordan–Wigner Mapping

Represent creation and annihilation operators as non-local combinations of qubit operators.

Jordan–Wigner mapping

First component: define naive per-site mapping

$$A^\dagger = \frac{1}{2}(X_p - iY_p)$$

$$\begin{aligned} A^\dagger|0\rangle &= |1\rangle \\ A^\dagger|1\rangle &= 0 \end{aligned}$$

$$A = \frac{1}{2}(X_p + iY_p)$$

$$A|0\rangle = 0$$

$$A|1\rangle = |0\rangle$$

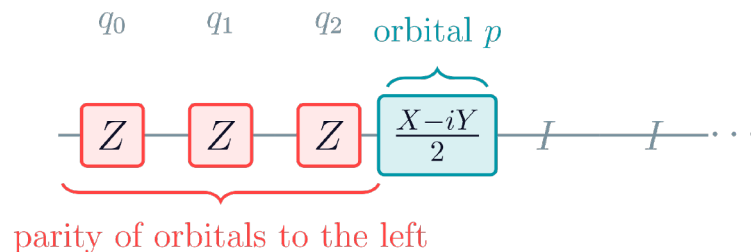
But still no anticommutation relation for different qubits p and q

$$A_p A_q = A_q A_p$$

Full solution: $a_p^\dagger \longmapsto \frac{1}{2}(X_p - iY_p) \otimes \bigotimes_{q < p} Z_q$

$$a_p \longmapsto \frac{1}{2}(X_p + iY_p) \otimes \bigotimes_{q < p} Z_q$$

Parity term enforces antisymmetry



Jordan & Wigner, Z. Phys. 47, 631 (1928)

H₂ under the Jordan–Wigner mapping

15 Pauli terms, coefficients in Ha

Pauli	coefficient
IIII	−0.09886397
ZIII	+0.17119775
IZII	+0.17119775
IIZI	−0.22278593
IIIZ	−0.22278593
ZZII	+0.16862219
ZIZI	+0.12054482
ZIIZ	+0.16586702

Pauli	coefficient
IZZI	+0.16586702
IZIZ	+0.12054482
IIZZ	+0.17434844
XXYY	−0.04532220
XYYX	+0.04532220
YXXY	+0.04532220
YYXX	−0.04532220

The diagonal part on its own determines the Hartree–Fock energy.
The entire correlation energy of H₂ lives in the XY terms.

Lowest eigenvalue of this 4-qubit operator:

−1.13727017 Ha

equal to FCI

Bravyi–Kitaev and parity mappings

There are other mappings with different ways to divide the labour between occupation and parity



mapping	occupation	parity
Jordan–Wigner	local	non-local
Parity	non-local	local
Bravyi–Kitaev	partial	partial

– BK stores partial sums on a binary tree, so both sets are $O(\log N)$

– Qubit tapering: another trick to be aware of, using symmetries to lower qubit overhead

Bravyi & Kitaev, Ann. Phys. 298, 210 (2002) and Seeley et al., JCP 137, 224109 (2012)

Bravyi, Gambetta, Mezzacapo & Temme, arXiv:1701.08213

What do we do with our qubit Hamiltonian?

We have H as a sum of Pauli strings. We want to find its smallest eigenvalue E_0 which is equal to the ground state energy.

1. Variationally minimise

$$E_0 \leq \frac{\langle \psi(\boldsymbol{\theta}) | \hat{H} | \psi(\boldsymbol{\theta}) \rangle}{\langle \psi(\boldsymbol{\theta}) | \psi(\boldsymbol{\theta}) \rangle}$$

Prepare a trial state, measure the energy, adjust and repeat.

→ **Variational quantum eigensolver (VQE)**

2. Encode the eigenvalue in a phase

$$e^{i\hat{H}t} |\phi_k\rangle = e^{iE_k t} |\phi_k\rangle$$

Run controlled time evolution and efficiently read out the phase.

→ **Quantum phase estimation (QPE)**

VQE: Variational Quantum Eigensolver

State Preparation

Prepare the trial state using a parameterised quantum circuit applied to a reference state (e.g. Hartree-Fock):

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta}) |\text{HF}\rangle$$

Circuit Ansatz Architecture

Multiple approaches to define the quantum circuit:

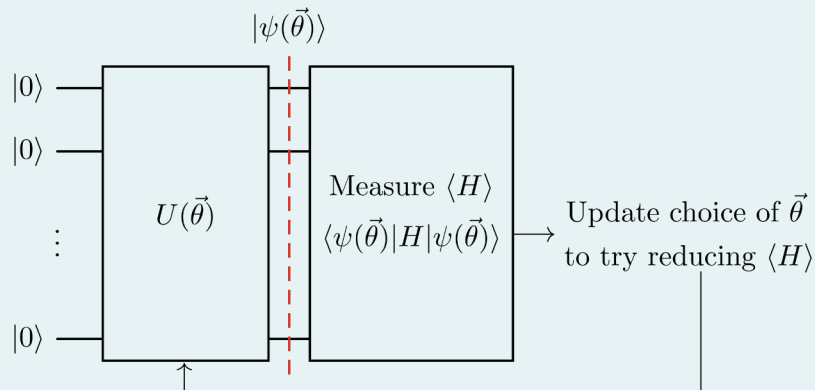
- Hardware-efficient approaches
- Chemistry-inspired (e.g., UCCSD)

$$\text{UCCSD: } U = e^{\hat{T}(\boldsymbol{\theta}) - \hat{T}^\dagger(\boldsymbol{\theta})}$$

Measurement & Optimisation

Measure expectation value of H and update circuit parameters $\boldsymbol{\theta}$ iteratively to lower energy further.

VQE circuit and optimisation loop



VQE problem 1: measurements

We have H as a sum of Pauli strings and could naively measure the expectation value of each Pauli string separately

$$\hat{H} = \sum_j c_j \hat{P}_j \quad \langle \hat{H} \rangle = \sum_j c_j \langle \hat{P}_j \rangle$$

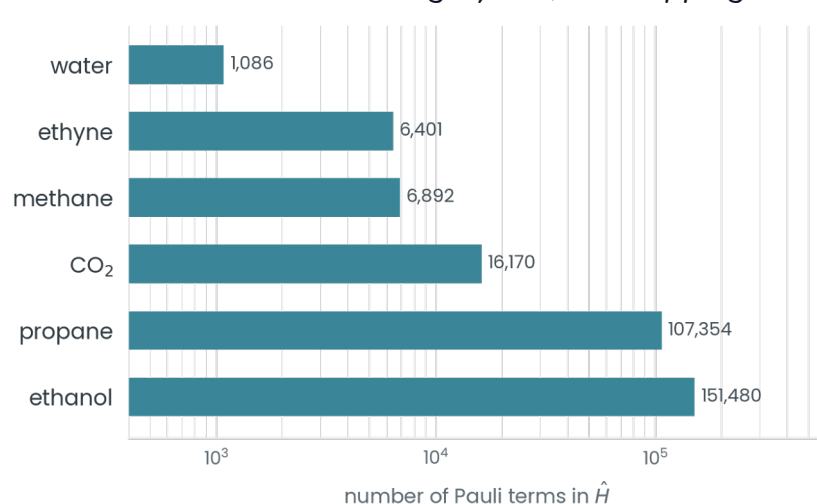
Can derive the number of measurements per Pauli and overall:

$$M_j \propto |c_j| \quad M \approx \frac{\lambda^2}{\epsilon^2}, \quad \lambda = \sum_j |c_j|$$

Lots of measurements needed, $O(\epsilon^{-2})$ is standard Monte Carlo scaling

In practice, the Pauli strings can be grouped into commuting terms that can be measured concurrently

STO-3G basis using PySCF, JW mapping

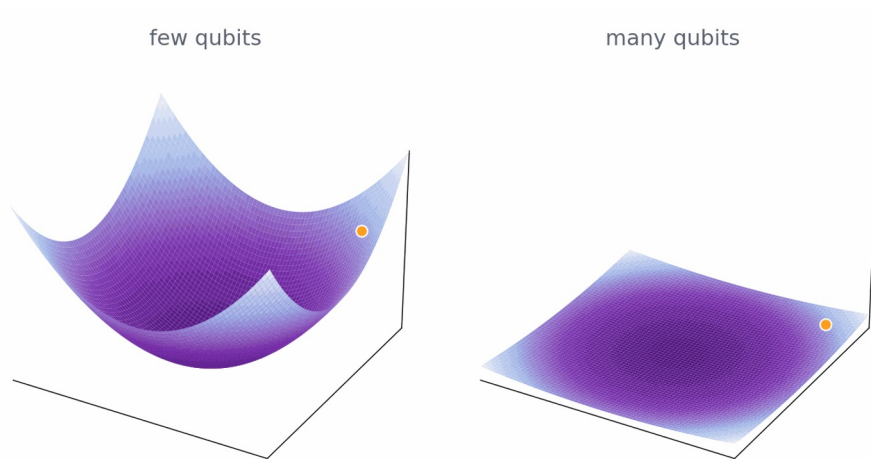


VQE problem 2: barren plateaux

For random parameterised circuits, gradients vanish exponentially with increasing qubits

$$\langle \partial_k E \rangle = 0, \quad \text{Var}[\partial_k E] \sim 2^{-n}$$

- Gradients are important to update parameters towards the ground state energy minimum
- If the gradient becomes exponentially small in qubit number, we don't know which direction to step!
- Barren plateaux don't *always* happen, e.g., some structured circuits or shallow circuits



McClean et al., *Nat. Commun.* 9, 4812 (2018)

VQE versus QPE

VQE

QPE (coming up)



Circuit depth

Variable, typically shallow for hardware

Deep, requires time evolution



Measurements

Many, related to Pauli decomposition

Few, depends on input state quality



Overall runtime

Standard quantum limit, ϵ^{-2}

Heisenberg scaling, ϵ^{-1}



Challenges

Unclear ansatz, barren plateaux

Time evolution is hard, state prep problem



Quantum phase estimation - setup

Eigenvalues of H become phases of a unitary

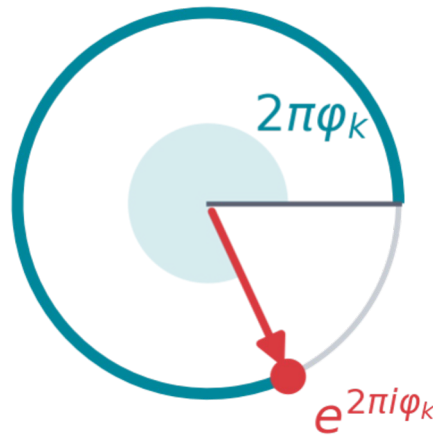
$$U = e^{i\hat{H}t} \implies U |\phi_k\rangle = e^{iE_k t} |\phi_k\rangle$$

First, some rearrangement: we focus on a phase φ_k between zero and one

$$e^{iE_k t} = e^{2\pi i \varphi_k}, \quad \varphi_k = \frac{E_k t}{2\pi} \bmod 1 \quad E_k = \frac{2\pi}{t} \varphi_k \bmod \frac{2\pi}{t}$$

Using QPE, we can obtain φ_k . But what value should we set for t ? Too large and φ_k will “loop around” as it is modulo 1, too small and our precision suffers.

Solution: renormalise H so that the eigenphases we care about are between zero and one - requires some approximations to the eigenspectrum of H .

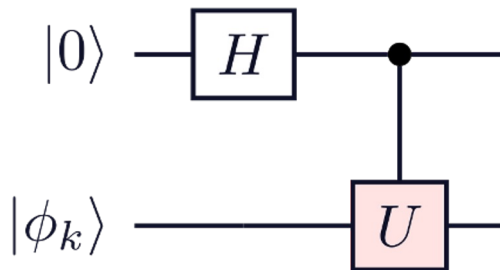


QPE machinery

Phase kickback - fundamental element of QPE

We can extract the phase from U onto an ancilla register (called “phase kickback”):

$$|+\rangle |\phi_k\rangle \xrightarrow{\text{controlled-}U} \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \varphi_k} |1\rangle \right) |\phi_k\rangle$$



Repeated phase kickback

Repeat the kickback with U , U^2 , U^4 , ... on m different ancillae. The phase φ_k is doubled each time:

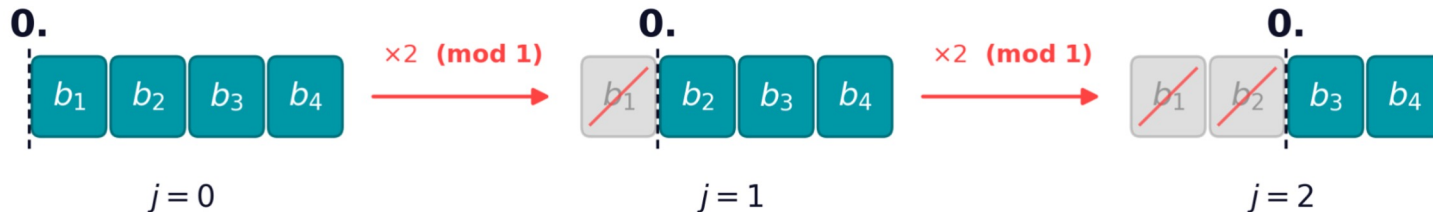
$$\frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 2^j \varphi_k} |1\rangle \right)$$

Suppose we represent the phase as a binary string:

$$\varphi_k = 0.b_1b_2\dots b_m = \sum_{i=1}^m b_i 2^{-i}$$

Doubling the phase causes us to lose the leading significant bit of precision:

$$2\varphi_k \bmod 1 = 0.b_2b_3\dots b_m$$

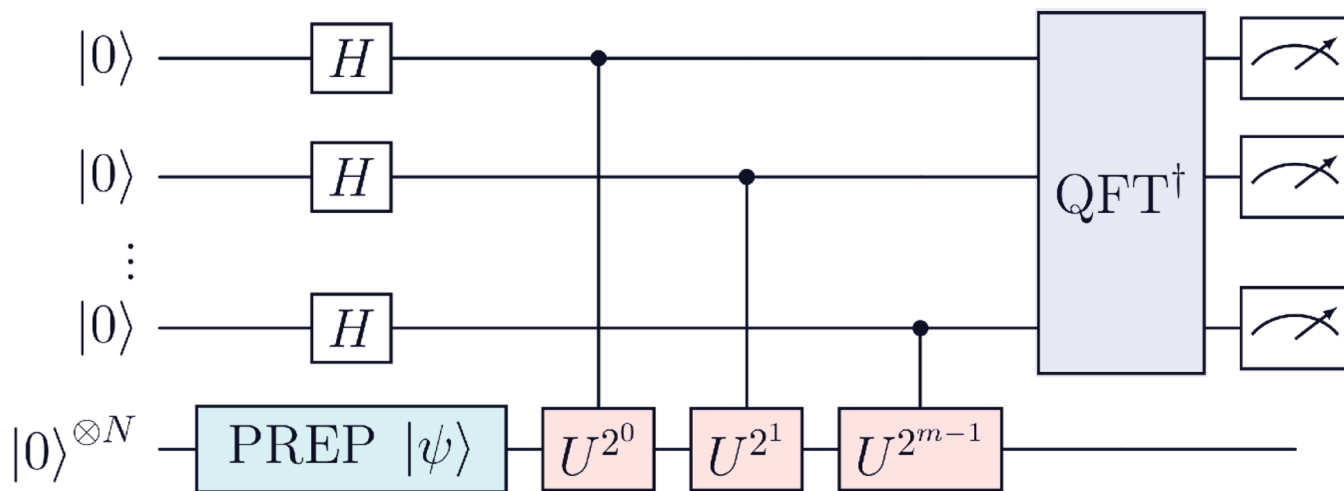


Read out bit-by-bit via the **inverse quantum Fourier transform**.

The full QPE circuit

Number of calls to controlled-U (runtime) $\sim \varepsilon^{-1}$

Number of additional ancillae $m \sim \log(\varepsilon^{-1})$



1 Initial state preparation

e.g., Hartree-Fock state
(not a focus of this tutorial)

2 Encoding into phases

Using kickback onto the ancillas

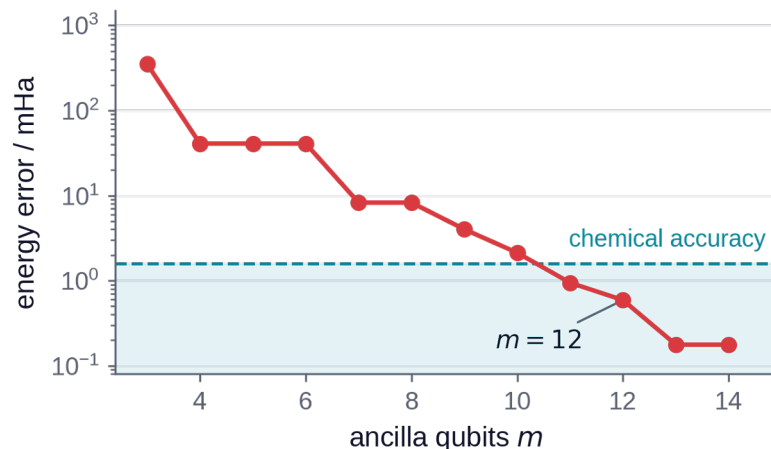
3 $\text{QFT}^\dagger$ and measure

phase encoded as binary string

Back to our hydrogen example

$U = \exp(iHt)$ with $t = 1$; exact $E_0 = -1.13727017$ Ha gives $\varphi_0 = 0.81899783$

m	Most likely bitstring	φ estimate	E / Ha	error / mHa
3	111	0.8750000	-0.7853982	351.9
4	1101	0.8125000	-1.1780972	-40.8
8	11010010	0.8203125	-1.1290099	8.3
10	1101000111	0.8193359	-1.1351458	2.1
12	110100011011	0.8190918	-1.1366798	0.6
14	11010001101010	0.8189697	-1.1374468	-0.2



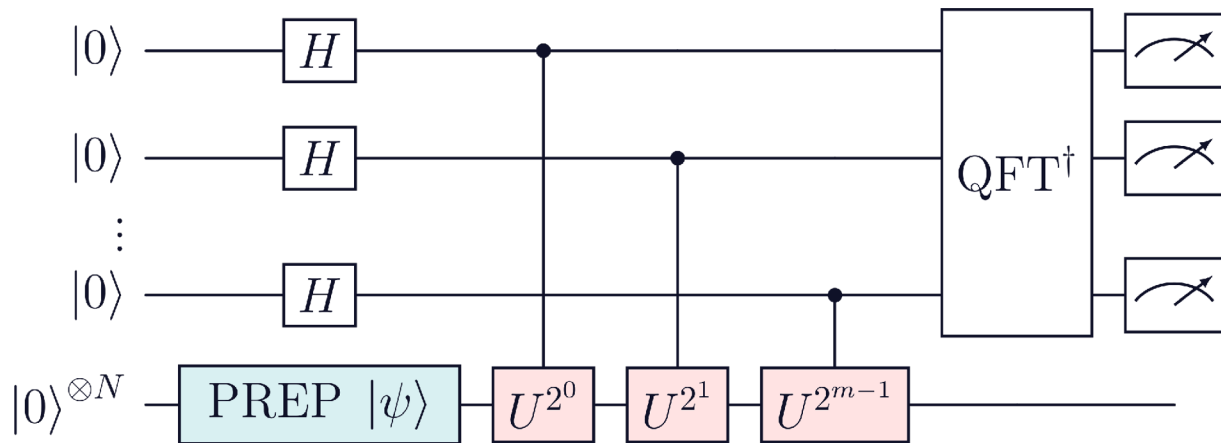
Observations

- Each extra qubit doubles the precision but also doubles the circuit depth!
- 10+ qubits needed for reasonable accuracy, requires thousands of controlled-U applications
- Each controlled-U operation will need thousands or more of hardware compatible gates

QPE has favourable scaling, but **needs fault tolerant hardware**

Costing up QPE

We want to break down the QPE algorithm into a circuit that can be run on hardware



$$\text{Cost}(\text{QPE}) \approx C(\text{state prep}) + O(1/\varepsilon) \times C(\text{controlled-}U) + C(\text{QFT}^\dagger)$$

Controlled- U breakdown typically dominates the gate count

We introduce the broad approaches in this session and go into greater detail after lunch

Trotterised Hamiltonian simulation

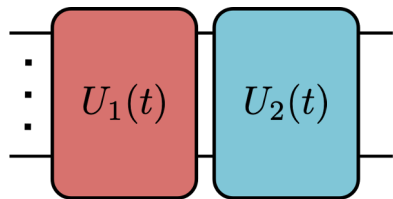
Starting from a Hamiltonian as a sum of Pauli strings, $H = \sum_{j=1}^N c_j P_j$, Trotterisation approximates the time propagator as

$$U(t) = e^{iHt} \approx \prod_{\alpha=1}^L e^{ia_{\alpha} P_{j_{\alpha}}}$$

Remember: we are dealing with matrices, only get equality when all P_j commute!

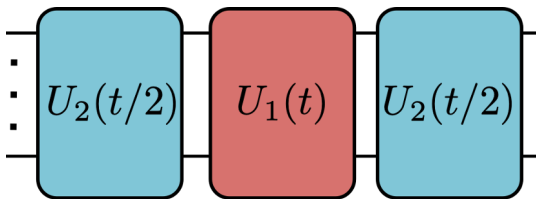
Lie-Trotter

$$\approx e^{ic_1 P_1 t} e^{ic_2 P_2 t}$$

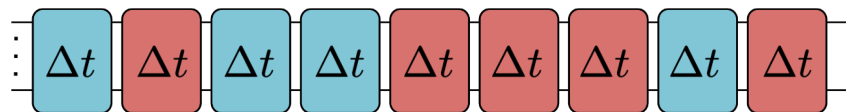


Suzuki-Trotter

$$\approx e^{ic_2 P_2 t/2} e^{ic_1 P_1 t} e^{ic_2 P_2 t/2}$$



randomised



Sample index i with probability $\propto c_i$

Block encodings

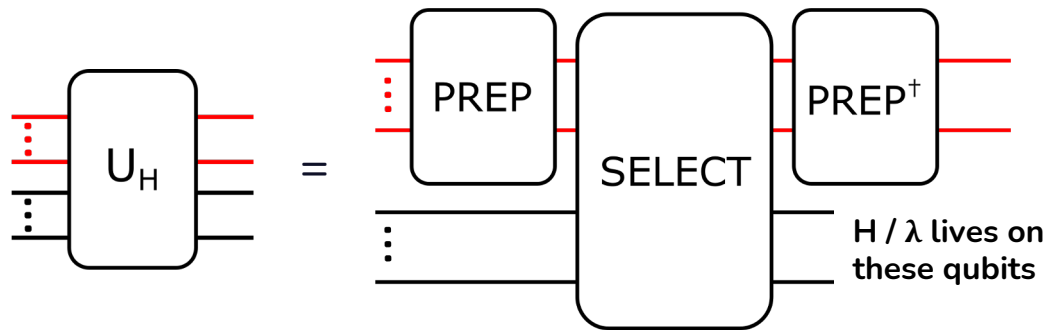
Quantum computers natively work with unitaries, but we can apply a general matrix if we can write it as a *linear combination of unitaries* (LCU). Pauli matrices are unitary, so our usual decomposition already does the job:

$$H = \sum_{j=1}^N c_j P_j$$

Using $\lceil \log_2 N \rceil$ ancilla qubits, the LCU is $U_H = \text{PREP}^\dagger \cdot \text{SELECT} \cdot \text{PREP}$, with

$$\text{PREP}|0_{\text{anc}}\rangle = \sum_{j=1}^N \sqrt{\frac{|c_j|}{\lambda}} |j\rangle$$

$$\text{SELECT} = \sum_{j=1}^N |j\rangle\langle j| \otimes P_j$$

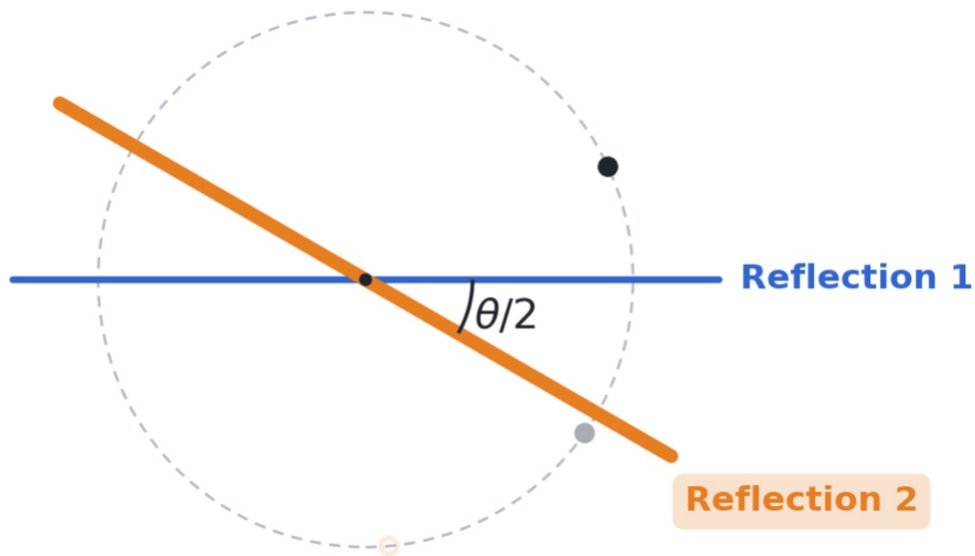


Gate breakdowns discussed next session!

Geometric aside: reflections and rotations

Two sequential reflections on a 2D plane separated by an angle $\theta / 2$.

The result is a rotation by θ .



Qubitisation

Apply our block encoding unitary to $|0\rangle |\phi_k\rangle$

$$U |0\rangle |\phi_k\rangle = \cos \theta_k |0\rangle |\phi_k\rangle + \sin \theta_k |\perp_k\rangle \quad \cos \theta_k = \frac{E_k}{\lambda}$$

We know that U is a reflection for an LCU of Pauli strings, so this is a reflection about $\theta_k / 2$

Now define another reflection along the axis of $|0\rangle |\phi_k\rangle$

$$R = 2 |0\rangle \langle 0|_a \otimes I - I$$

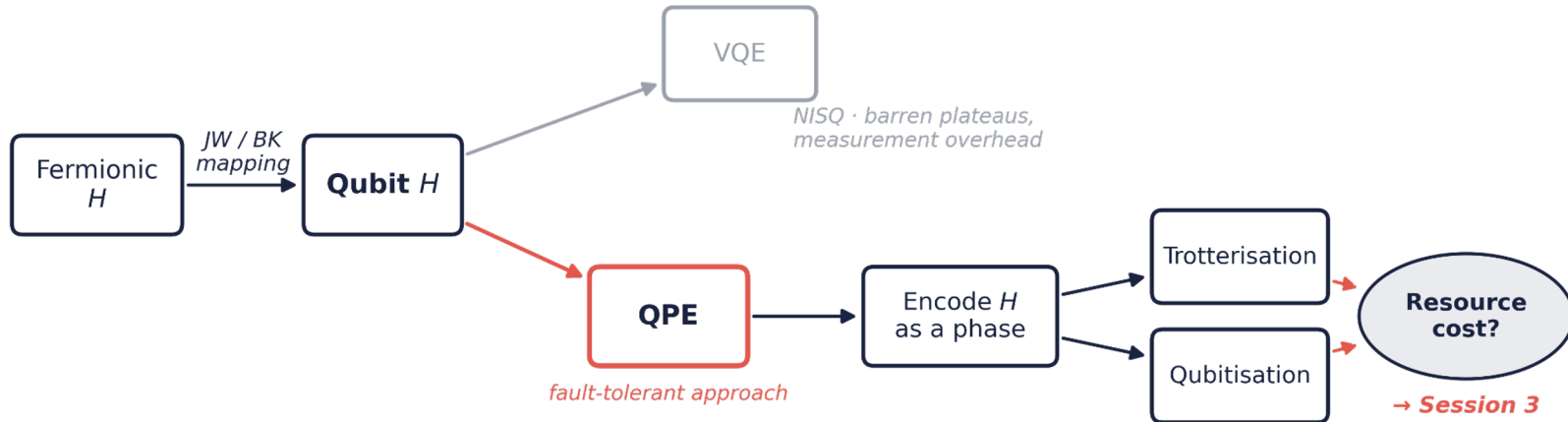
Together, we arrive at the walk operator W , which applies a rotation of θ_k

$$W = RU$$

We can use QPE with the controlled walk operator!

Trotter vs qubitisation comparison coming up later

Wrapping up



Next up:

- But how much does this all cost? We want to use our quantum resources as efficiently as possible and compare resource costs of different approaches.
- We'll go through gate breakdowns on paper and in software!

Advanced topics to read up on

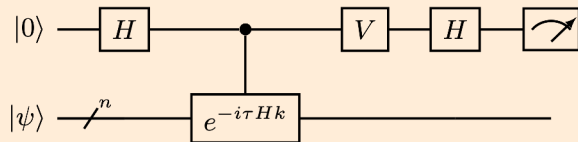
Lee et al., PRX Quantum 2, 030305 (2021)

$$h_{ijkl} = \sum_{\mu, \nu=1}^M \chi_i^{(\mu)} \chi_j^{(\mu)} \zeta_{\mu\nu} \chi_k^{(\nu)} \chi_l^{(\nu)}$$

Hamiltonian factorisation

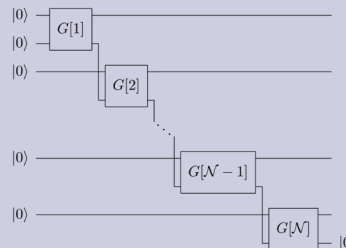
Rewrite fermionic Hamiltonian to lower QPE overhead

Blunt et al., PRX Quantum 4, 040341 (2023)



Alternative QPE methods

Trade-off between ancillas, depth, and shots



Berry et al., PRX Quantum 6, 020327 (2025)

State preparation techniques

Beyond-HF techniques like DMRG to lower state prep cost

Georges et al., npj Quantum Information 11, 55 (2025)

$$\hat{H}_{\text{LCU},1} = \sum_{p,q=0}^{D-1} \omega_{pq} (-i)^{\mu(p,q)} \sum_{j=0}^{N-1} \prod_{k=0}^{M-1} X_{jM+k}^{p_k} Z_{jM+k}^{q_k} i^{p_k \wedge q_k}$$

First-quantisation

Potential to make savings for large basis sets



Thanks! Questions and discussion?

